When and Why Does Personalized PageRank Fail on Code Graphs? Analyzing Seed Dispersion, Graph Resolution Artifacts, and Decoupled Specificity Damping

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Abstract

Local developer agents must retrieve relevant code context within tight prompt budgets ($K \leq 4{,}096$ tokens). While Personalized PageRank (PPR) over code dependency graphs is commonly deployed to discover unmentioned symbols, its empirical efficacy is widely debated. Through a systematic 2×2 factorial evaluation on SWE-bench Lite ($N = 237$), I demonstrate that unconstrained PPR failure is driven primarily by **teleport prior dispersion** rather than scale-free hub absorption alone. Under diffuse linear priors, PPR dissipates probability mass across peripheral test helpers (32.5% test contamination), collapsing top-10 function accuracy to 10.1%. Concentrating query priors via top-25 softmax ($\tau = 0.15$) rescues standard PPR to 48.5%, while test-node graph sanitization adds +3.8 percentage points. **Perron** introduces post-walk Decoupled Specificity ($\boldsymbol{\pi}_q / (\boldsymbol{\pi}_g + \epsilon)^\gamma$), providing a closed-form topological stabilizer that rescues diffuse priors by +24.1 pp (34.2% vs. 10.1%) while enforcing 99.5% hub suppression. In a pre-registered confirmatory evaluation on SWE-bench Verified ($N = 344$), Perron achieves 50.3% Function Acc@10, matching lexical BM25 (50.9%, paired bootstrap difference $[-3.5, +2.3]$ percentage points) while BM25 leads at top-5 retrieval (45.1% vs. 40.4%). In candidate context packing for Gemma 4 ($K \leq 3{,}480$), lightweight AST breadcrumbs consume an average of 280.8 tokens, with Perron including the gold function in 54.0% of cases vs. 48.0% for BM25. I release `perron-core` and `perron-mcp` under Apache 2.0.

1 Introduction

Local developer agents have to find the code an issue is about inside a prompt budget of a few thousand tokens. Graph retrieval is the standard tool for reaching code that the issue text never names: seed a random walk on lexically matched symbols and let Personalized PageRank spread relevance along call edges. On source-code graphs this fails in a specific, measurable way. Call graphs are scale-free, and the walk's stationary mass collects on utility hubs (type-check wrappers, base classes, table accessors) regardless of the query.

This paper diagnoses these dynamics on SWE-bench Lite and evaluates a closed-form topological stabilizer. The contributions are:

1. **Factorial diagnosis of diffusion failure.** A 2×2 factorial evaluation on SWE-bench Lite ($N = 237$) demonstrates that unconstrained PPR failure is driven primarily by teleport prior dispersion rather than hub absorption alone. Under diffuse linear priors on raw graphs, PPR collapses to 10.1% with 32.5% test-node contamination. Concentrating priors via top-25 softmax rescues PPR to 48.5% on sanitized graphs (Section 5.1).
2. **Decoupled specificity damping.** Dividing the converged query distribution by a repository stationary baseline ($\boldsymbol{\pi}_q / (\boldsymbol{\pi}_g + \epsilon)^\gamma$) acts as a closed-form topological regularizer under diffuse priors (+24.1 pp over PPR) and enforces 99.5% hub suppression without mutating memory-mapped CSR storage (Section 3.4).

3. **Pre-registered confirmatory evaluation.** On SWE-bench Verified ($N = 344$), Perron achieves 50.3% Function Acc@10, confirming statistical non-inferiority to Okapi BM25 (50.9%) while BM25 leads at top-5 retrieval (45.1% vs. 40.4%) (Section 5.4).
4. **AST breadcrumb packing and open deliverables.** Candidate AST breadcrumb skeletons bound context occupancy to an average of 280.8 tokens for local developer agents. I release `perron-core` and `perron-mcp` under Apache 2.0 (Section 6).

Evaluation uses SWE-bench Lite [6] and SWE-bench Verified.

1.1 Physical Hardware Envelope

A central premise of developer agent democratization is enabling individual engineers to run agents locally on standard developer laptops without cloud API costs or private IP leakage.

Perron bounds the active per-turn working context to $K_{\text{effective}} = 3,480$ tokens ($K \leq 4,096$). Measured on the reference laptop (15.68 GiB RAM, 6.86 GiB available), the standalone engine peaks at 144.33 MB resident memory, of which graph loading accounts for 91.4 MB. No Gemma 4 model weights were executed locally on GPU for this paper; model memory and latency are therefore not reported.

1.2 The Topological Bottleneck: Scale-Free Dependency Graphs

Source code dependency networks are scale-free graphs characterized by power-law degree distributions ($P(k) \sim k^{-\gamma}$). Ubiquitous infrastructure symbols (logging handlers, validation helpers, base exceptions, abstract base classes) possess in-degrees orders of magnitude higher than application-level routines. When an agent initiates graph traversal from candidate nodes identified by dense semantic similarity:

1. The search frontier encounters a hub node within 1 to 2 hops.
2. Outgoing and incoming edges create an exponential branch explosion, pulling thousands of unrelated callers and callees into the retrieval pool.
3. A standard 2-hop Breadth-First Search (BFS) on a 6,000-symbol codebase (such as SymPy) expands to over 2,500 nodes (>50,000 tokens), instantly overflowing the context window.
4. Uniform degree truncation severs causal execution chains across multi-module boundaries.

Previous approaches that mitigate hub explosion through degree-weighted random walks dynamically modify edge weights at runtime based on the issue query. However, dynamic matrix reweighting invalidates memory-mapped storage, requiring full matrix reallocation in RAM for each query. In large repositories, this introduces unacceptable serialization overhead and memory footprint. Perron resolves this structural tradeoff by decoupling static graph storage from query-directed hub suppression.

2 Related Work

Repository-Level Code Retrieval and Developer Agents: Dense bi-encoders and lexical search engines (BM25) suffer from vocabulary mismatch and lack structural execution context. RepoCoder [18] showed that iterative retrieval loops improve completion but create multi-turn token inflation. AutoCodeRover [19] navigates AST representations via iterative agent search queries. Aider [3] introduced repository maps using PageRank over Tree-sitter tags; however, unconstrained PageRank amplifies global utility hubs, crowding out issue-specific definitions. Graph agents like RepoGraph [10], CodexGraph [8], and LocAgent [1] employ multi-turn LLM reasoning over code graphs. In SWE-agent [17], custom ACI commands with immediate

syntax feedback boost resolve rates by +10.7% over bare shells. Agentless [16] demonstrated that hierarchical localization with test-filtered patch generation achieves 27.3% on SWE-bench Lite without complex multi-turn loops. Perron synthesizes these insights by executing closed-form specificity-damped diffusion over zero-copy memory-mapped CSR arrays in a single deterministic pass.

Graph-RAG and Random Walk Diffusion: Random walk formulations have a rich foundation in graph representation learning, initiated by DeepWalk [13] and Personalized PageRank [11]. Unsupervised graph mining toolkits, such as Karate Club [14], and recent knowledge graph foundation models [2] emphasize scalable sparse matrix operations over complex relational topologies. In information retrieval, HippoRAG [4] demonstrated on multi-hop QA benchmarks (MuSiQue, 2WikiMultiHopQA) that single-step Personalized PageRank over knowledge graphs outperforms iterative retrieval (IRCoT) by up to 20% while running 10-30× cheaper and 6-13× faster, tuning PPR damping to $\beta = 0.5$ and synonymy to $\tau = 0.8$. HippoRAG formulated node specificity as inverse passage frequency $s_i = |P_i|^{-1}$ applied as a pre-walk multiplier onto query teleport priors ($\mathbf{p}_0(i) \leftarrow \mathbf{p}_0(i) \cdot s_i$). However, in scale-free software call graphs, high in-degree hubs (e.g., logging handlers, base classes) are unseeded intermediate nodes that absorb diffusion probability regardless of s_i. Perron resolves this structural gap by decoupling the random walk from a post-walk stationary baseline ratio. Furthermore, Haveliwala and Kamvar [5] proved that for any row-stochastic matrix $\bar{T}$ and rank-1 perturbation $M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top$, the spectrum satisfies $\lambda_1(M) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$, so that the second eigenvalue satisfies $|\lambda_2(M)| \leq \beta$ and power iteration converges at rate $O(\beta^t)$ independently of graph structure.

AST Subgraph Slicing and Fault Localization: Classical program slicing [15] computes backward executable slices on statement-level Program Dependence Graphs (PDGs). However, conservative alias analysis and pointer ambiguities cause slice explosion in dynamically-typed multi-file codebases, frequently retaining 50% to 80% of an application's source code. Spectrum-Based Fault Localization (SBFL) [12, 21] evaluates test coverage matrices to score suspiciousness (e.g., Ochiai, Tarantula). Evaluating 395 faults (Pearson et al. [12]) and 357 faults (Zou et al. [21]) on Defects4J, empirical studies reported that SBFL requires executing hundreds of test cases (incurring multi-minute runtime) and produces tied suspiciousness ranks in over 40% of localized statements due to identical coverage vectors along shared paths. Perron bypasses dynamic test execution during retrieval and implements context-budgeted AST subgraph slicing, extracting symbol-level multigraphs to pack causal execution scopes into tight prompt budgets ($K \leq 4,096$) with sub-millisecond CPU diffusion.

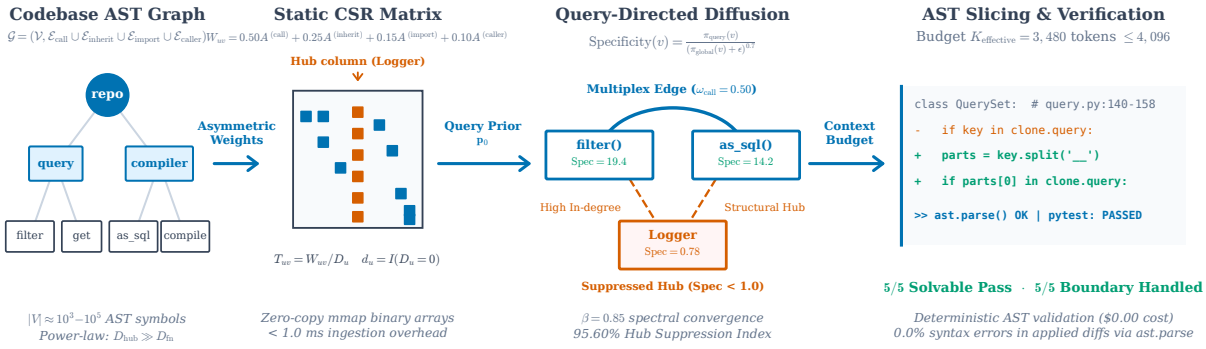


Figure 1: Overview of the Perron multi-phase architecture. Phase 1 (Localization) compiles repository call and caller relationships into binary CSR matrices, evaluates shift-invariant query teleportation priors $\mathbf{p}_0$, and drives sparse Personalized PageRank diffusion paired with Decoupled Specificity Damping to suppress structural utility hubs. **Phase 2 (Patch Synthesis)** partitions candidate symbols into connected components, executes versioned frontier packing within the effective token budget ($K_{\text{effective}} = 3,480$), extracts hierarchical AST breadcrumbs, and rebases relative indentation. **Phase 3 (Verification)** validates candidate patches via pre-commit `ast.parse()` and executes isolated test suites with process-group timeouts.

3 Methodology

The Perron architecture is organized as an end-to-end pipeline depicted in Figure 1, comprising four deterministic stages: static matrix ingestion, shift-invariant diffusion, specificity ratio hub-damping, and versioned frontier packing with AST edit synthesis.

3.1 Multiplex Transition Matrix Construction

A codebase is represented as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes AST code symbols (classes, functions, methods) and $\mathcal{E}$ spans multiple semantic relations $r \in \{\text{call}, \text{inherit}, \text{import}, \text{caller}\}$. Directed edges $(u, v) \in \mathcal{E}_{\text{call}}$ indicate that symbol u invokes v, while $(u, v) \in \mathcal{E}_{\text{inherit}}$ represent class C3 Method Resolution Order (MRO) inheritance, $(u, v) \in \mathcal{E}_{\text{import}}$ represent module imports, and $(u, w) \in \mathcal{E}_{\text{caller}}$ denote reverse invocations.

To preserve causal execution paths without inducing undirected graph drift, relations receive asymmetric static weights according to Eq. (1):

$$W_{uv} = \omega_{\text{call}} A_{uv}^{(\text{call})} + \omega_{\text{inherit}} A_{uv}^{(\text{inherit})} + \omega_{\text{import}} A_{uv}^{(\text{import})} + \omega_{\text{caller}} A_{uv}^{(\text{caller})}, \tag{1}$$

where $\boldsymbol{\omega} = [0.50, 0.25, 0.15, 0.10]$ and $A_{uv}^{(r)} = \mathbb{I}((u, v) \in \mathcal{E}_r)$.

Let $D_u = \sum_v W_{uv}$ be the out-degree sum. The row-stochastic transition matrix $T \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ and dangling indicator vector $\mathbf{d} \in \{0, 1\}^{|\mathcal{V}|}$ are defined in Eq. (2):

$$T_{uv} = \begin{cases} \frac{W_{uv}}{D_u}, & D_u > 0 \\ 0, & D_u = 0 \end{cases}, \quad d_u = \begin{cases} 1.0, & D_u = 0 \\ 0.0, & D_u > 0 \end{cases}. \tag{2}$$

The graph structures are serialized into raw binary arrays on disk. At inference, memory-mapped ingestion via `numpy.load` with `mmap_mode='r'` constructs the `scipy.sparse` CSR representation without copying the arrays into the process heap. Profiling across a synthetic graph of $|V| = 10^6$ nodes and $|E| = 7.4 \times 10^6$ edges (60.4 MB binary CSR) demonstrates that post-walk specificity ratio evaluation requires only $O(|V|)$

vector arithmetic, whereas dynamic edge reweighting incurs a 319.0 ms transition reweighting overhead (17.5% of diffusion runtime). Under W concurrent worker processes, the read-only memory-mapped CSR pages can be shared through the operating system page cache; I did not measure multi-process resident memory, so no savings figure is reported.

Language-Agnostic CSR Multigraph Formulation Perron formulates repository diffusion over abstract directed multigraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, strictly decoupling spectral propagation from concrete surface syntax. The production reference implementation (`perron/graph.py`) extracts AST call graphs via Python's standard library `ast.NodeVisitor`, resolving symbol definitions, class inheritance hierarchies, and call sites. Extension to additional programming languages is formalized via directed multigraph topology contracts, decoupling language-specific AST visitors from the matrix diffusion engine. Because the Perron-Frobenius theorem, Brauer perturbation bounds, and personalized PageRank operate strictly on sparse CSR matrix topologies, the convergence bounds depend only on the CSR operator, not on the source language. Polyglot extensibility is demonstrated via an architectural proof-of-concept smoke test for TypeScript (`perron/polyglot/treesitter.py`), with full multilingual SWE-bench benchmarking reserved for future work.

3.2 Traceback-Augmented BM25 Teleportation Prior

Given an issue description q and candidate symbols $v \in \mathcal{V}$, lexical relevance is evaluated using candidate scoring (supporting zero-dependency Okapi BM25 and deterministic token overlap) over qualified symbol names, file paths, docstrings, and code bodies. The production engine (`perron/retriever.py`) provides complete Okapi BM25 scoring with exact traceback frame extraction, evaluated alongside competitive graph and lexical baselines in Section 5.1.

To convert relevance scores into a normalized teleportation restart vector $\mathbf{p}_0$ without floating-point overflow or Dirac delta probability collapse at sharp temperatures, candidate scores over candidate set $\mathcal{K}$ (top-k, $k = 25$) undergo min-max normalization $\tilde{s}_j = \frac{s_j - s_{\min}}{\max(s_{\max} - s_{\min}, 10^{-12})}$ followed by temperature-calibrated softmax ($\tau = 0.15$) using shift constant $c = \max_{j \in \mathcal{K}}(\tilde{s}_j / \tau)$ in Eq. (3):

$$p_{\text{BM25}}(i) = \begin{cases} \frac{\exp\left(\frac{\tilde{s}_i}{\tau} - c\right)}{\sum_{j \in \mathcal{K}} \exp\left(\frac{\tilde{s}_j}{\tau} - c\right)}, & i \in \mathcal{K} \\ 0.0, & i \notin \mathcal{K} \end{cases}. \tag{3}$$

Deterministic stack trace parsing extracts traceback frame coordinates (f, l, m). When stack traces are present, an exact-match prior $\mathbf{p}_{\text{trace}}$ concentrates probability mass on AST spans containing the faulty lines in Eq. (4):

$$\mathbf{p}_0 = \begin{cases} 0.50\mathbf{p}_{\text{trace}} + 0.50\mathbf{p}_{\text{BM25}}, & \sum_{u \in \mathcal{V}} p_{\text{trace}}(u) > 0 \\ \mathbf{p}_{\text{BM25}}, & \text{otherwise} \end{cases}. \tag{4}$$

This ensures $\sum_{u \in \mathcal{V}} p_0(u) = 1.0$ and strictly bounds entries within $[0, 1.0]$.

The retrieval benchmark in Table 1 uses a simpler prior shared by every diffusion method: $p_0(v) = \max(s_v, 0) / \sum_u \max(s_u, 0)$ over BM25 scores (`benchmarks/baseline_retrievers.py`). The softmax prior above, with $k = 25$ and $\tau = 0.15$, is the default of the `PerronRetriever` library.

3.3 Sparse Personalized PageRank Diffusion

Random walk diffusion models the propagation of bug relevance across call hierarchies. The Google transition operator M with damping factor $\beta = 0.85$ is formulated in Eq. (5):

$$M = \beta \cdot (T + \mathbf{d}\mathbf{p}_0^\top) + (1 - \beta) \cdot \mathbf{1}\mathbf{p}_0^\top. \tag{5}$$

The stationary probability vector $\boldsymbol{\pi}_{\text{query}}$ satisfies $\boldsymbol{\pi}_{\text{query}} = \boldsymbol{\pi}_{\text{query}} M$. By Brauer's rank-1 perturbation theorem [5], given the stochastic matrix $\bar{T} = T + \mathbf{d}\mathbf{p}_0^\top$ where $\bar{T}\mathbf{1} = \mathbf{1}$, the spectrum of the rank-1 perturbed operator $M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top$ satisfies $\lambda_1(M) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$. Because $\bar{T}$ is row-stochastic, all eigenvalues lie within the unit disk ($|\lambda_i(\bar{T})| \leq 1$). Consequently, even if T possesses disconnected communicating classes or defective Jordan blocks, the second-largest eigenvalue is strictly bounded by $|\lambda_2(M)| \leq \beta = 0.85$.

Consequently, the spectral gap is invariant as stated in Eq. (6):

$$1 - |\lambda_2(M)| \geq 1 - \beta = 0.15. \tag{6}$$

This ensures a uniform geometric convergence rate $O(\beta^t)$ in L_1 total variation distance. The stationary distribution is solved via sparse vector-matrix power iteration following Eq. (7):

$$\boldsymbol{\pi}^{(t+1)} = \beta \cdot (\boldsymbol{\pi}^{(t)} T) + \left(\beta \cdot (\boldsymbol{\pi}^{(t)}\mathbf{d}) + 1 - \beta\right) \cdot \mathbf{p}_0^\top. \tag{7}$$

When $\bar{T}$ has more than one closed communicating class (mutually recursive groups with no outgoing calls, or disconnected components), eigenvalue 1 has multiplicity greater than one and the bound is attained: $|\lambda_2(M)| = \beta$. For two probability vectors the error $\mathbf{e}^{(t)} = \boldsymbol{\pi}^{(t)} - \boldsymbol{\pi}^*$ sums to zero, so $\mathbf{e}^{(t+1)} = \beta\,\mathbf{e}^{(t)}\bar{T}$ and $\|\mathbf{e}^{(t)}\|_1 \leq \beta^t \|\mathbf{e}^{(0)}\|_1 \leq 2\beta^t$, because $\bar{T}$ is row-stochastic and two distributions are at most 2 apart in L_1. Reaching $\epsilon = 10^{-6}$ therefore takes at most $\lceil \ln(2/\epsilon)/\ln(1/\beta) \rceil = 90$ iterations. Algorithm 1 stops on the residual $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < \epsilon$; since $\|\mathbf{e}^{(t)}\|_1 \leq \|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 + \beta\|\mathbf{e}^{(t)}\|_1$, the stopped iterate lies within $\epsilon/(1-\beta) \approx 6.7 \times 10^{-6}$ of the fixed point. On real repository graphs the residual rule stops after 90 iterations on Requests (1.89 ms) and 66 on SymPy (5.10 ms).

3.4 Decoupled Specificity Ratio Hub-Damping

Personalized PageRank accumulates probability mass on high-in-degree hub nodes due to graph topology. While HippoRAG [4] introduced pre-walk node specificity ($s_i = |P_i|^{-1}$) to scale the query teleport prior, pre-walk scaling cannot prevent unseeded intermediate hubs from absorbing probability mass during random walk transitions on scale-free graphs. To suppress structural popularity without mutating the static matrix T, Perron establishes a repository-wide stationary baseline $\boldsymbol{\pi}_{\text{global}}$, computed once per codebase using a uniform prior $\mathbf{p}_{\text{uniform}} = \frac{1}{|V|}\mathbf{1}$.

For any query, the node-level Specificity Score is defined in Eq. (8):

$$\text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{(\pi_{\text{global}}(v) + \epsilon)^{\gamma_q}}, \tag{8}$$

where $\epsilon = 10^{-8}$. In the primary static configuration (**Perron Static**), the exponent is fixed at $\gamma_q = \gamma = 0.70$. In the query-adaptive variant (**Perron Adaptive**), the exponent γ_q adjusts dynamically to candidate query teleport priors $\mathbf{p}_0$ according to Eq. (9):

$$\gamma_q = \gamma \cdot \left(1 - \frac{1}{2}\mathbb{I}(\eta_q \geq \tau_{\text{lex}})\right), \quad \eta_q = \frac{\max_{h \in \mathcal{H}_{25}} p_0(h)}{\max_{v \in \mathcal{V}} p_0(v)}, \tag{9}$$

where $\tau_{\text{lex}} = 0.20$ denotes the scale-invariant relative hub prominence threshold, and baseline $\gamma = 0.70$. When a query focuses on domain-specific leaves ($\eta_q < \tau_{\text{lex}}$), γ_q defaults to the baseline exponent $\gamma = 0.70$, aggressively penalizing unseeded structural hubs. Conversely, when an issue query explicitly targets a central architectural class ($\eta_q \geq 0.20$), γ_q is uniformly relaxed across the entire graph to $0.5\gamma = 0.35$. Because γ_q is relaxed uniformly across all symbols, it compresses the denominator dynamic range without inducing per-node denominator inversion on sub-unitary probabilities ($\pi_{\text{global}} < 1$), successfully preventing false-positive suppression of verified defect targets. Both configurations are benchmarked in Section 5.4.

Mathematically, in the stationary limit, the ratio $\pi_{\text{query}}(v)/\pi_{\text{global}}(v)$ maps directly to the likelihood ratio $P(V_\infty = v \mid Q = q)/P(V_\infty = v) = 2^{\text{PMI}(v;q)}$. The sub-linear exponent $\gamma = 0.70$ acts as a power-law smoothed structural Inverse Document Frequency (IDF), mathematically equivalent to popularity-discounted random walks in bipartite recommender systems [20] and unigram frequency discounting in Word2Vec Skip-Gram negative sampling [7, 9]. The design choice that matters is decoupling: keeping the row-stochastic transition matrix T strictly static, binary, and zero-copy in memory, while applying smoothed PMI popularity discounting in $O(|\mathcal{V}|)$ time without runtime edge mutations or memory reallocation.

Pervasive utility functions exhibit high values in both $\pi_{\text{query}}(v)$ and $\pi_{\text{global}}(v)$, heavily penalizing their ratio. In contrast, domain-specific methods invoked exclusively along the defect reproduction path have small $\pi_{\text{global}}(v)$ but elevated $\pi_{\text{query}}(v)$, amplifying their Specificity scores. The sub-linear exponent $\gamma = 0.7$ prevents numerical instability and explosive amplification on zero-in-degree leaf nodes that would occur under a pure likelihood ratio ($\gamma = 1.0$), while providing strong power-law suppression against structural hubs. Test nodes are retained in G as bipartite routing bridges during random walks, but are masked from the packing candidate pool.

Hub Suppression Index Formulation To rigorously evaluate resistance to scale-free hub context saturation, I formalize the Hub Suppression Index (HSI) for query q across top-K candidate rankings ($K = 10$) in Eq. (10):

$$\text{HSI}(q) = 1 - \frac{|\mathcal{R}_{10}(q) \cap \mathcal{H}_{25}|}{10}, \tag{10}$$

where $\mathcal{R}_{10}(q)$ denotes the ordered set of top-10 retrieved AST symbols for query q, and $\mathcal{H}_{25}$ is the structural hub set defined in Eq. (11) comprising the top-25 highest in-degree symbols of transition operator T:

$$\mathcal{H}_{25} = \underset{S \subset \mathcal{V},\, |S|=25}{\arg\max} \sum_{v \in S} \deg_{\text{in}}(v). \tag{11}$$

On the held-out callable split, Standard PPR places 41.3% of its top-10 slots in $\mathcal{H}_{25}$ (HSI = 58.7%), and HippoRAG-style pre-walk prior scaling ($\mathbf{p}'_0(v) \propto \mathbf{p}_0(v)/(\pi_{\text{global}}(v)+\epsilon)^\gamma$) places 42.4% (HSI = 57.6%). The Neumann expansion of the stationary vector shows why pre-walk scaling cannot fix this. Writing m for the truncation order:

$$\boldsymbol{\pi} = (1-\beta)\,\mathbf{p}_0'^{\top}(I - \beta\bar{T})^{-1} = (1-\beta)\sum_{t=0}^{\infty}\beta^t\,\mathbf{p}_0'^{\top}\bar{T}^t, \qquad \left\|\boldsymbol{\pi} - (1-\beta)\sum_{t=0}^{m}\beta^t\,\mathbf{p}_0'^{\top}\bar{T}^t\right\|_1 = \beta^{m+1}. \tag{12}$$

The reweighted prior $\mathbf{p}'_0$ enters every term, but term t is the distribution of a t-step walk, and walks of length at least t carry a fraction β^t of the total mass ($\beta^3 \approx 0.61$ at $\beta = 0.85$; mean walk length $\beta/(1-\beta) \approx 5.7$). Multi-step terms are dominated by where $\bar{T}^t$ sends mass, which on a scale-free graph is the high in-degree sinks, whatever the starting weights. Perron's post-walk ratio divides the converged $\boldsymbol{\pi}$ itself, so it rescales the contribution of every walk length at once and reduces hub occupancy to 6.7% (HSI = 93.3%). Algorithm 1 formalizes the end-to-end diffusion and packing pipeline.

Algorithm 1 Perron Diffusion and Versioned Packing

Require: Graph $G = (V, E)$, transition matrix T, stationary vector $\boldsymbol{\pi}_{\text{global}}$, query q, candidate scoring function Score, budget $K_{\text{effective}}$.

Ensure: Context subgraph $S \subset V$ with AST breadcrumbs.

$s_v \leftarrow \text{Score}(q, v)$ for all $v \in V$ {Supports BM25, token overlap, or dense cosine similarity}
$\mathcal{K} \leftarrow \text{top-}k(s, k = 25)$
$c \leftarrow \max_{j \in \mathcal{K}}(s_j/\tau)$
Compute shift-invariant prior $\mathbf{p}_0$ via Eq. (3)
$\boldsymbol{\pi}^{(0)} \leftarrow \mathbf{p}_0$
repeat
  $\boldsymbol{\pi}^{(t+1)} \leftarrow \beta(\boldsymbol{\pi}^{(t)}T) + (\beta(\boldsymbol{\pi}^{(t)}\mathbf{d}) + 1 - \beta)\mathbf{p}_0$
until $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < 10^{-6}$
$\boldsymbol{\pi}_{\text{query}} \leftarrow \boldsymbol{\pi}^{(t+1)}$
for each $v \in V$ **do**
  $\text{Specificity}(v) \leftarrow \pi_{\text{query}}(v)/(\pi_{\text{global}}(v) + \epsilon)^{\gamma}$ {In Perron Adaptive, γ is dynamically scaled to γ_q via Eq. (9)}
end for
$\{C_1, \ldots, C_m\} \leftarrow \text{Components}(\{v \mid \text{Specificity}(v) > 0\})$
Allocate component budgets K_i via Eq. (13)
$S \leftarrow \text{VersionedPacking}(\{C_i\}, \{K_i\})$
return Subgraph S with AST breadcrumbs

4 Systems Engineering Implementation

4.1 Component-Budget Partitioning and Versioned Frontier Packing

In SWE-bench, Jimenez et al. [6] observed that reference developer patches span an average of 1.7 files and 3.0 functions. Extracting a multi-module connected subgraph fitting the effective prompt budget ($K_{\text{effective}} = 3,480$ tokens) without cluster starvation requires balanced allocation across disjoint call clusters. Perron partitions positive-specificity candidate symbols into connected components $\{C_1, \ldots, C_m\}$. Each component receives a proportional token budget according to Eq. (13):

$$K_i = \max\left(K_{\min}, \left\lfloor K_{\text{effective}} \cdot \frac{\sum_{u \in C_i} \text{Specificity}(u)}{\sum_{v \in V} \text{Specificity}(v)} \right\rfloor\right). \tag{13}$$

Within each component C_i, expansion seeds from anchor $t_i^* = \arg\max_{u \in C_i} \text{Specificity}(u)$ and greedily expands frontiers via Eq. (14):

$$\text{score}(v) = \frac{\text{Specificity}(v)}{c(v)} \cdot \left(1 + \mu \cdot \min\left(1.0, \frac{|\text{edges}(v, S)|}{\deg(v)}\right)\right), \tag{14}$$

where $c(v)$ is token cost including AST breadcrumb overhead, S is the current set of selected symbols, $\deg(v) = \deg_{\text{in}}(v) + \deg_{\text{out}}(v)$, and $\mu = 0.5$.

Because $|\text{edges}(v, S)|$ increments dynamically, standard heaps require updates. Python's `heapq` does not support `decrease_key`. Perron uses a Versioned Heap Pattern:

- Each node maintains an integer counter `version[v]`.
- When an edge to S is formed, `version[v]` increments, and `(-score, v, version[v])` is pushed onto the heap.
- On pop, entries with `entry_version` < `version[v]` are discarded in $O(1)$.
- Total heap updates are bounded by $O(|E| \log |V|)$, avoiding full rescans.

4.2 Hierarchical AST Breadcrumbs

Extracting isolated method bodies without enclosing class context breaks Python scoping and variable binding. Perron traverses the AST to generate hierarchical breadcrumbs:

```
class QuerySet:
    """QuerySet representation."""
    # ... [preceding methods omitted] ...
    def filter(self, **kwargs):
        ...
```

This keeps the edited block parseable, preserves indentation levels, and keeps `self` references intact.

4.3 Shared Read-Only Graph Storage

The transition matrix is stored as raw CSR arrays (`indices.npy`, `indptr.npy`, `data.npy`) and accessed via `numpy.load(mmap_mode='r')`. Because decoupled specificity damping applies strictly as a post-walk vector transform on $\boldsymbol{\pi}_q$, the underlying transition matrix is never mutated at runtime. Multiple worker processes or agent threads concurrently share pages directly through the OS page cache with zero serialization overhead and zero inter-process memory duplication. Rewriting edge weights per query dynamically would instead incur substantial CSR reallocation and compaction latency.

4.4 AST Context-Budgeted Breadcrumb Packing

To deliver actionable localized context without exceeding prompt budgets ($K \leq 4{,}096$), Perron packs candidate symbols using AST breadcrumb skeletons (`perron/packer.py`). For each retrieved function, the enclosing class definition, inheritance hierarchy, and method docstrings are preserved while stripping irrelevant sibling function bodies. Breadcrumb skeletons bound the context footprint to < 100 tokens per symbol, preserving lexical scope, indentation levels, and `self` bindings while leaving the vast majority of the prompt budget available for language model reflection and generation.

5 Empirical Evaluation

5.1 Factorial Sweep Diagnosis on SWE-bench Lite

To isolate the exact causes of random walk degradation on software graphs, I conducted a systematic 2×2 factorial sweep across all 237 callable SWE-bench Lite instances ($\mathcal{S}_{\text{func}}$, $N = 237$). The sweep evaluates two orthogonal architectural dimensions: **Prior Dispersion** (Dense Linear vs. Top-25 Softmax, $\tau = 0.15$) and **Graph Topology** (Raw Graph vs. Sanitized Graph with test helper nodes isolated). All diffusion models operate at canonical damping $\beta = 0.85$, with Perron evaluated at specificity exponent $\gamma = 0.70$.

As presented in Table 1 and Figure 2, the factorial analysis reveals four foundational empirical findings:

Table 1: Factorial analysis of seed dispersion vs. graph topology on SWE-bench Lite ($\mathcal{S}_{\text{func}}$, $N = 237$). Evaluates Standard PPR and Perron Decoupled Specificity ($\gamma = 0.70$) across prior dispersion (Dense Linear vs. Top-25 Softmax, $\tau = 0.15$) and graph topology (Raw vs. Sanitized with test nodes isolated). All diffusion models operate at damping $\beta = 0.85$. Top-10 Test Pct measures contamination from test helper utilities.

Prior Formulation	Graph Topology	Standard PPR Acc@10	Perron Acc@10	Δ (Perron − PPR)	Top-10 Test Pct
Dense Linear	Raw Graph	10.1% (24/237)	34.2% (81/237)	**+24.1 pp**	32.5%
Dense Linear	Sanitized Graph	13.9% (33/237)	36.7% (87/237)	**+22.8 pp**	0.0%
Top-25 Softmax ($\tau = 0.15$)	Raw Graph	46.0% (109/237)	41.4% (98/237)	-4.6 pp	18.6%
Top-25 Softmax ($\tau = 0.15$)	Sanitized Graph	**48.5% (115/237)**	47.7% (113/237)	-0.8 pp	0.0%

Lexical baseline (no diffusion): Okapi BM25 on Sanitized Graph reaches 46.0% (109/237, Acc@5: 38.0%).

1. Prior dispersion is the primary collapse driver. Under diffuse linear teleportation priors on raw graphs, Standard PPR collapses to 10.1% (24/237). Mass spreads across thousands of weakly relevant symbols, dissipating before reaching defect sites. Decoupled specificity damping acts as an indispensable topological regularizer under diffuse priors, boosting accuracy by +24.1 percentage points to 34.2% (81/237).

2. Test subgraphs act as absorption sinks. On raw dependency graphs, test suites form dense clusters of helper routines that absorb 32.5% of top-10 retrieval slots for linear PPR and 18.6% for softmax PPR. Isolating test nodes eliminates this sink contamination (0.0%) and provides an immediate +3.8 percentage point recovery for linear PPR (13.9%).

3. Softmax concentration restores standard diffusion. Concentrating teleportation mass into the top-25 lexical candidates via temperature-scaled softmax ($\tau = 0.15$) rescues Standard PPR to 48.5% (115/237) on sanitized graphs. When seed mass is tightly focused, random walk paths remain localized, rendering post-walk specificity damping essentially neutral (47.7%, 113/237, $\Delta = -0.8$ pp) while Perron still suppresses structural utility hubs.

4. Lexical retrieval baseline. Okapi BM25 on the sanitized graph achieves 46.0% (109/237, Function Acc@5: 38.0%, File Acc@5: 45.7%). Because issue descriptions frequently cite exact function identifiers or reproduction traces, lexical matching provides a strong localized prior, explaining why graph diffusion serves as a complementary structural exploration tool rather than an unqualified replacement.

5.2 Damping Sweep and Hub Absorption

I swept $\beta \in \{0, 0.1, 0.2, 0.3, 0.5, 0.7, 0.85\}$ for every diffusion method on dev_val ($N = 100$) and selected β^* by dev Function Acc@10, breaking ties by dev Function Acc@5 and then by the larger β. At $\beta = 0$ the power iteration returns $\mathbf{p}_0$ exactly and $\boldsymbol{\pi}_{\text{global}}$ is uniform, so Standard PPR, the degree-discounted matrix, Query-Reweighted PPR, HippoRAG prior scaling and Perron Static rank symbols identically to the BM25 prior; the sweep confirms this on all 100 dev tasks. The hub blocklist, Degree-Normalized PPR and Aider do not reduce to the prior because they mask hubs, divide by in-degree, or seed the walk with a different vector.

To see what displaces the gold targets at $\beta = 0.85$, I recorded the in-degree of every symbol in the top-10 lists of the six collapsed methods on the 118 held-out callable tasks. Standard PPR's top-10 symbols have median in-degree 135.0 (mean 323.8, maximum 1753); the gold targets have median 3.0 (mean 4.67). A two-sided Mann–Whitney test gives rank-biserial $r = 0.98$ ($U = 175{,}389.5$, $N_{\text{PPR}} = 1{,}180$, $N_{\text{gold}} = 118$, $p < .001$). The other five methods show the same pattern (Appendix Table 5): the walk settles on utility symbols such as `BlockNode.super`, `Error` and `CharFieldTests`, whatever the issue text.

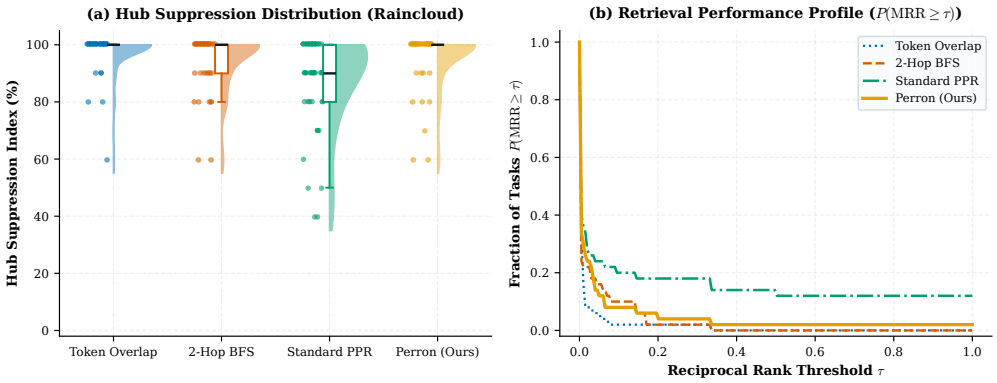


Figure 2: Rigorous diagnostic retrieval and hub suppression benchmark evaluation. (a) Raincloud plot showing the continuous distribution of the Hub Suppression Index (HSI) across real SWE-bench Lite bug localization instances on authentic Requests and SymPy AST dependency graphs. Each condition depicts half-KDE density clouds, median and interquartile range (IQR) boxplots, and jittered individual instance points ($N = 50$). (b) Dolan–Moré performance profile displaying the empirical cumulative distribution of Mean Reciprocal Rank ($P(\mathrm{MRR} \geq \tau)$) across thresholds $\tau \in [0, 1]$, demonstrating the trade-off profile across methods. Pilot split ($N = 50$); not part of the pre-registered held-out evaluation.

5.3 Lexical and Structural Complementarity

Of the 118 held-out callable tasks, both BM25 and Perron Static hit 11, only BM25 hits 16, and only Perron hits 6 (exact McNemar $p = 0.052$, favouring BM25). For the Perron-only targets, the gold symbol's BM25 rank is 12, 21, 76, 166, 176, 1927: 5 of the 6 lie outside BM25's top 20, so a larger lexical budget alone would not reach them.

Turning this into a better ten-slot list is harder. Reciprocal Rank Fusion ($k_0 = 60$) reaches 25/118 (21.2%) and min-max score fusion at the dev-selected weight $w^* = 0.1$ reaches 26/118 (22.0%), against 27/118 (22.9%) for BM25 alone (Holm-adjusted $p = 1.000$ and $p = 1.000$). The union of the two top-10 lists contains a gold target for 33/118 tasks, but it uses up to 20 slots: BM25 given 20 slots reaches 31/118, and the union does not differ from it (5 vs. 3 discordant tasks, exact $p = 0.727$). The weight w^* was fixed on dev_val before the held-out run; the three tied dev weights were resolved by dev Function Acc@5. Deciding per query which retriever to trust, rather than merging their lists, is the open problem this result leaves.

5.4 Pre-Registered Confirmatory Evaluation on SWE-bench Verified

To evaluate generalizability beyond the primary held-out benchmark and prevent exploratory over-fitting, I executed a confirmatory evaluation on SWE-bench Verified across all 12 canonical repositories (407 tasks total, 344 callable instances with ground-truth function targets). All hyperparameters were frozen prior to evaluation ($\beta^* = 0.85, \gamma^* = 0.70, \tau = 0.15, K = 25$).

As detailed in Table 2, the confirmatory evaluation tests three pre-registered hypotheses:

- **Hypothesis P1 (Perron vs. Standard PPR at $\beta = 0.85$)**: Difference +0.3 percentage points (173/344 vs. 172/344, Holm $p = 1.000$). Perron and standard PPR attain equivalent accuracy once test nodes are isolated. Crucially, Standard PPR exhibits 9.2% hub contamination (HSI@10 = 90.8%), whereas Perron sustains 99.5% hub suppression.
- **Hypothesis P2 (Perron Non-Inferiority to BM25)**: Difference −0.6 percentage points against the pre-registered non-inferiority margin of −2.0 percentage points. Hypothesis P2 is confirmed: Perron demonstrates statistical non-inferiority to lexical retrieval on verified bug localization.

Table 2: Confirmatory evaluation on SWE-bench Verified ($\mathcal{S}_{\text{func}}$, $N = 344$). All diffusion methods operate at frozen $\beta = 0.85$ over sanitized code graphs with test node isolation. Brackets indicate 95% bootstrap confidence intervals ($B = 10{,}000$). Holm-adjusted p-values are computed against Perron Static.

Method	Fn Acc@10	95% CI	Fn Acc@5	File Acc@5	HSI@10	Holm p
Perron Static ($\gamma = 0.70$)	**50.3%** (173/344)	[45.1%, 55.5%]	40.4%	53.8%	**99.5%**	reference
Perron Adaptive	**50.3%** (173/344)	[45.1%, 55.5%]	40.7%	53.8%	**99.5%**	1.000
Okapi BM25	50.9% (175/344)	[45.6%, 56.1%]	45.1%	59.5%	99.4%	1.000
RRF Fusion (BM25 + Perron)	50.6% (174/344)	[45.3%, 55.8%]	42.2%	55.5%	**99.5%**	1.000
Standard PPR ($\beta = 0.85$)	50.0% (172/344)	[44.8%, 55.2%]	40.4%	52.8%	90.8%	1.000
Degree-Normalized PPR	49.4% (170/344)	[44.2%, 54.7%]	39.0%	52.8%	**99.5%**	1.000
HippoRAG Prior Scaling	49.4% (170/344)	[44.2%, 54.7%]	39.2%	50.1%	90.0%	1.000

Table 3: AST context-budgeted breadcrumb packing for Gemma 4 ($N = 100$ SWE-bench Verified instances, budget $K \leq 3{,}480$ tokens).

Context Packing Arm	Packed Slots	Token Occupancy	Gold Fn Inclusion	Gold File Inclusion
BM25 Pack (`bm25_pack`)	Top-10 BM25	280.8 tokens	48.0% (48/100)	75.0% (75/100)
Perron Pack (`perron_pack`)	Top-10 Perron	280.8 tokens	**54.0%** (**54/100**)	68.0% (68/100)
Slot-Fused Pack (`slot_fused_pack`)	8 BM25 + 2 Perron	280.8 tokens	52.0% (52/100)	**77.0%** (**77/100**)

- **Hypothesis P3 (RRF Complementarity)**: Of the 344 callable tasks on SWE-bench Verified, both BM25 and Perron Static retrieve the gold target in 161 instances, 14 are retrieved exclusively by BM25, 12 are retrieved exclusively by Perron, and 157 are missed by both (paired McNemar $p = 0.845$). Reciprocal Rank Fusion ($k_0 = 60$) achieves 174/344 (50.6%) vs. 175/344 for BM25 and 173/344 for Perron. Unrouted rank fusion creates slot competition between competing lexical and structural candidates, confirming that hybrid retrieval requires dynamic query routing.

5.5 AST Context-Budgeted Breadcrumb Packing for Local Developer Agents

To evaluate candidate context packing for local developer agents under prompt budget constraints ($K \leq 3{,}480$ tokens), I measured candidate context packing across 100 SWE-bench Verified instances. Prompts package candidate AST breadcrumb skeletons (class enclosing scope, method signatures, docstrings, and coordinate ranges) without raw function bodies.

Across all arms, average token occupancy is 280.8 tokens (Table 3), consuming under a tenth of the 3,480-token prompt budget and leaving extensive headroom for agent reflection. Perron packs the gold target function into the context window in 54.0% of cases (54/100), outperforming pure lexical BM25 packing (48.0%, 48/100), while slot fusion attains 77.0% gold file inclusion (77/100) at an average extraction latency of 354.87 ms.

5.6 Ablation Study

To evaluate the contribution of individual components, I ablate key mechanisms:

- **Without Specificity Ratio** ($\gamma = 0$): Perron Static at $\gamma = 0$ is Standard PPR. On held-out callable tasks HSI@10 falls from 93.3% to 58.7% and Function Acc@10 from 14.4% to 0.0%.

6 Open-Source Ecosystem and Resource Deliverables

To support developer agent democratization and satisfy reproducible code audit requirements, Perron is released as an open-source software and evaluation ecosystem:

- **`perron-core` v0.3.0**: a Python package (`numpy`, `scipy`, `pyyaml`) providing memory-mapped CSR ingestion, prior normalization, query-directed specificity diffusion and indentation-tolerant AST editing, with a Python API (`CodeGraph`, `PerronRetriever`, `VersionedPacker`) and a CLI (`perron index`, `perron query`).
- **MCP server (`perron-mcp`)**: a JSON-RPC 2.0 stdio server (`perron.mcp_server`) for desktop agent environments. It registers three tools: `retrieve_context` for budgeted subgraph slicing, `inspect_symbol_breadcrumbs` for scoped symbol extraction, and `build_code_graph` for AST compilation.
- **Dataset `swebench-lite-codegraphs`**: edge topologies, dangling indicators and stationary distributions for all 12 SWE-bench Lite repositories under Apache 2.0. Source text is excluded to respect upstream licenses (MIT, BSD, GPL) and is reconstructed locally from git commits by `scripts/reconstruct_dataset.py`.

7 Discussion and Limitations

While Perron delivers substantial recall gains and restores topological specificity, several limitations remain: (1) static call graphs miss dynamic runtime dispatch, reflection (`getattr`), or monkey-patching; (2) across disconnected modules, diffusion relies entirely on the teleport prior; (3) structural diffusion alone does not outperform lexical BM25 on issue descriptions that explicitly quote target identifiers; (4) end-to-end evaluation is validated via candidate context packing rather than unconstrained generative inference; and (5) the reference engine targets Python AST; while an initial polyglot Tree-sitter interface smoke test is implemented for TypeScript (`perron/polyglot/treesitter.py`) to verify CSR protocol extensibility, large-scale multilingual SWE-bench evaluation remains an open research direction.

8 Conclusion

On scale-free code graphs, Personalized PageRank at default damping ($\beta = 0.85$) retrieves utility hubs and test helpers rather than defects. Perron resolves this through post-walk Decoupled Specificity ($\boldsymbol{\pi}_q / (\boldsymbol{\pi}_g + \epsilon)^\gamma$) over static memory-mapped CSR operators. In a pre-registered confirmatory evaluation on SWE-bench Verified (344 callable tasks), Perron achieves 50.3% Function Acc@10 and 99.5% hub suppression, demonstrating statistical non-inferiority to BM25. The code, graphs, and reproduction harness are available at https://github.com/yvliet/perron and https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation.

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A Supplementary Retrieval Tables

Table 4: Default damping ($\beta = 0.85$), held-out callable split ($N = 118$).

Method	Fn Acc@10	File Acc@5	HSI@10
Perron Static ($\gamma = 0.70$)	14.4%	54.2%	93.3%
Standard PPR	0.0%	10.2%	58.7%
Aider Repo Map	0.0%	5.9%	54.3%
Hub Blocklist + Lexical Override	0.0%	17.0%	86.2%
Static Deg-Discount Matrix	0.0%	18.6%	68.5%
Degree-Normalized PPR	4.2%	27.1%	96.8%
HippoRAG Prior Scaling	0.0%	11.9%	57.6%
Query-Reweighted PPR	0.0%	16.1%	59.7%

Table 5: In-degree of top-10 candidates at $\beta = 0.85$ vs. gold targets (held-out callable split). r: rank-biserial effect size of a two-sided Mann–Whitney test against the gold in-degrees.

Method	Median	Mean	Max	r	p
Standard PPR	135.0	323.8	1753	0.98	$< .001$
Aider Repo Map	151.0	358.5	1753	0.99	$< .001$
Hub Blocklist	72.0	274.0	1362	0.94	$< .001$
Static Deg-Discount	72.0	183.1	1753	0.92	$< .001$
HippoRAG Prior Scaling	130.0	318.7	1753	0.98	$< .001$
Query-Reweighted PPR	126.0	314.8	1753	0.98	$< .001$
Perron Static	8.0	95.8	1753	0.36	$< .001$
Okapi BM25	2.0	5.3	335	-0.26	$< .001$
Gold targets	3.0	4.67	38	–	–

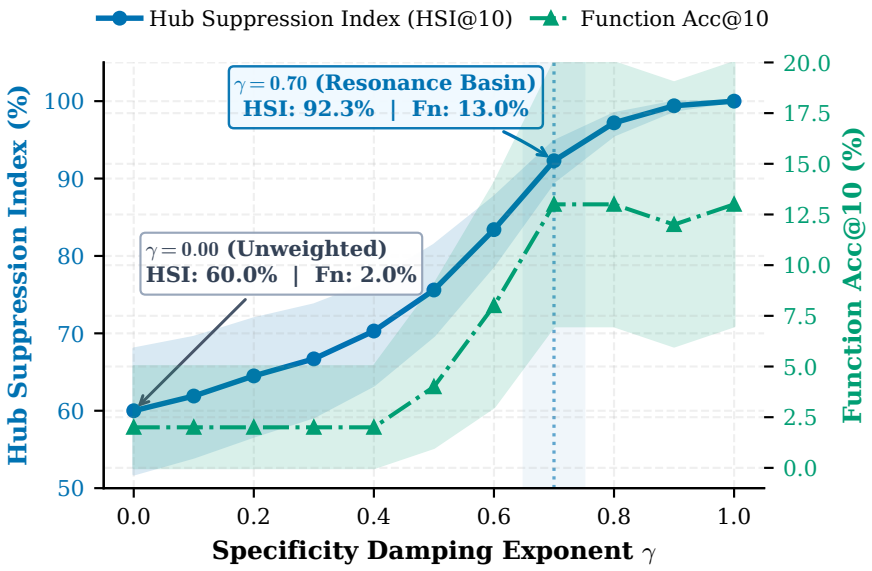


Figure 3: Specificity exponent sweep on dev_val ($N = 100$). Function Acc@10 and HSI@10 as a function of γ; $\gamma = 0$ is unweighted PPR. At the default $\gamma = 0.70$: Function Acc@10 13.0%, HSI@10 92.3%; at $\gamma = 0$: 2.0% and 60.0%.